TECHNISCHE UNIVERSITÄT WIEN

institute of telecommunications

Machine Learning Laboratory

389.247

Gerald Matz, Dimitrios Kalodikis

Institute of Telecommunications

Technische Universität Wien

Vienna, Austria

Winter Semester 26/27

Probabilistic Approaches:

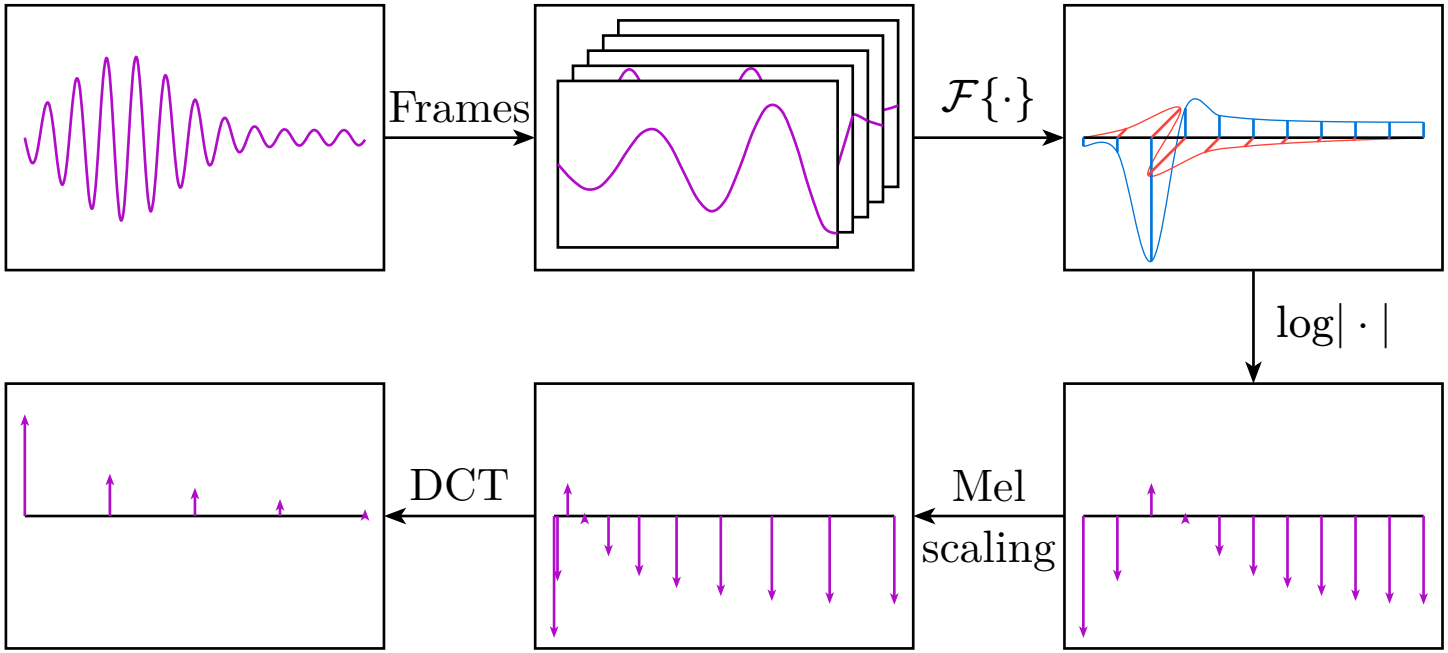
An application

Application: Speech Recognition

- Speech is sequence of phonemes (sounds)
- Some phonemes can have arbitrary length
- Pitch and volume do not determine phoneme (high/low, loud/soft voice)
- Human hearing perception is
  - logarithmical in pitch and volume
  - indifferent to phase
- Feature extraction: Mel Frequency Cepstral Coefficients (MFCC)

$\Rightarrow$ Example of **feature engineering**

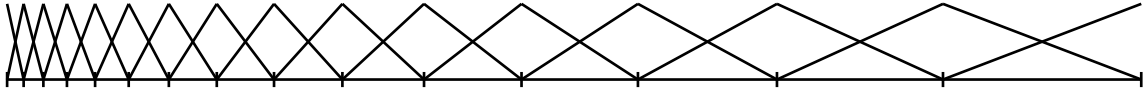
Application: Speech Recognition (II)



Application: Speech Recognition (III)

Typical values:

- 25ms frames with 10ms overlap
- Hamming windows
- Mel scale: $m = 1127 \cdot \ln\left(\frac{f}{700} + 1\right)$
- 40 coefficients per frame
- Downsampling to Mel coefficients via triangular overlapping windows:



Side fact: Convolutions (filtering) in the time domain become additive in the MFCC domain

Application: Speech Recognition (IV)

Idea of Markov chain based speech recognition:

- Markov model describes phonemes
- Phonemes can be stretched over multiple frames
- MFCC are Gaussian emissions with mean and covariance according to current phoneme
- Parameters of the model can be learned using EM and labelled data
- Inference can be done using Viterbi

Shortcomings of the basic approach:

- Phoneme duration modelled as exponential distribution
- Gaussian emission is to simplistic ($\Rightarrow$ Gaussian mixtures)
- Phoneme sounds depend on context (neighbouring phonemes)